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POWER MODULE PRODUCT CATALOG

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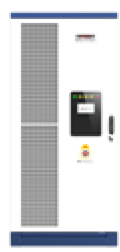

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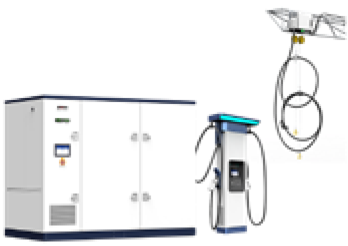

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V2G Charging Solution

EXP30K2 Fast DC Wallbox Charger

30kW DC Charging Power with Two DC Charging Connectors + (o) 22kW AC Type 2 Charging Socket/Cable

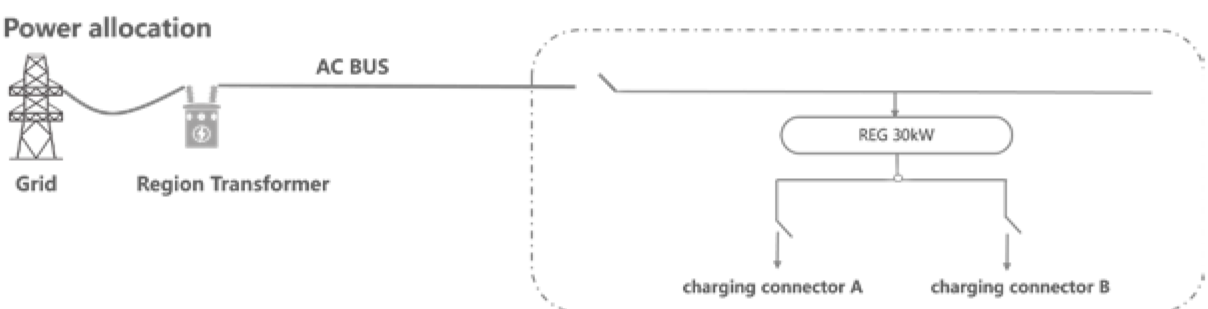
Introduction

EXP30K2 DC charger is specially designed with wall mounted, pedestal and movable installations and fully compatible with CCS/NACS/CHAdeMO/GBT EV charging standard. It features small size, light weight and easy to install and move, user friendly interface, IP54 protection, robust and durable design for various outdoor applications. It can deliver up to 30kW charging power and 150V to 1000V charging voltage. It can be engineered with optional 22kW/32A AC Type 2 charging socket/cable and portable accessory.

Main Features

- Flexible installation scenarios: wall mounted/ pedestal installation/portable accessory
- CE/RED, CB, TUV US/CA, Energy Star and OCA certification
- WEB/APP Maintenance Backend System enhances the operation and maintenance effectiveness
- AC Type 2 charging socket/cable configuration support
- Optional cable retractor, credit card (POS), VIN code support
- CCS1, CCS2, GBT, CHAdeMO and NACS support
- BABA BABA support

Application



Technical Parameters

Input Parameters

AC frequency	45-65Hz
AC voltage	CE: 3 phase+N+PE, 380/400Vac±10% UL: 3 phase+PE, 480Vac±10%
Power factor	>0.99
Total harmonic current	<3% (rated input)

Output Parameters

Output DC voltage range	DC 150V-1000V
Max DC output current	100A
Max DC output power	30kW
Constant power voltage range	300~1000V
Max system efficiency	>95% (rated input and output)
Optional AC charging output	AC charging socket/cable 32A/3P/22kW

Working Environment

Ambient temperature	-30~+65°C, full power output below 45°C, power derating 5%/°C above 45°C
Storage temperature	-40~+75°C
Working humidity	0~95%
Working altitude	2000 m

Standard

Backend	OCPP 1.6J, support firmware update to 2.0x, OTA support, P&C/smart charge/power management support
CCS PLC communication	DIN70121 and ISO15118
CHAdeMO	CHAdeMO V1.2
GBT	GB/T 18487.1-2015, GB/T 27930-2015, GB/T 20234.1~3-2015
Certification	TUV US+CA, Energy Star, TUV CE, RED, CB, OCA UL 2202/CSA C22.2, EN IEC61851-21-2 Class B; EN IEC61851-1/EN IEC61851-23/EN61851-24
Protection class	IP54/IK10

Function and Interface

HMI	Tempered glass protective 7" TFT touch screen LCD, three RGB LED, EPO
Metering	DC energy meter, DL/T645 standard, optional MID meter
Monitor interface	LAN and LTE wireless modem support
Charging mode	Auto basis, time basis, amount basis, energy basis, SOC basis
Power switch mode	Two DC charging connectors charging in serial, AC socket/cable charging in parallel
Payment	RFID, Mobile APP, POS (opt)
MBE function support	WEB side Maintenance Backend System enhances the operation and maintenance effectiveness

Mechanical

Dimensions (W*H*D mm)	610 * 610 * 270 mm
Weight (kg)	62 kg (excluding power module, power module is 19kg)

Charging Protection

Overcurrent protection, short circuit protection, overvoltage protection, undervoltage protection

Insulation monitoring, protect earth connect monitoring

Reverse polarity protection, overtemperature protection

RCDB for AC charging socket/cable configuration protection

Configuration

Charger module configuration	REG1K0100G/U 30kW1000V charger power module
Charging connector configuration	One or two CCS/NACS/GBT connectors configuration, CHAdeMO + CCS/NACS/GBT connectors configuration, optional AC Type 2 charging socket/cable
Charging connector capacity	GBT: 125A CCS: 125A NACS: 125A CHAdeMO: 125A AC: 32A/22kW
Optional pedestal	Pedestal with two cable retractors and the light bar
Optional portable accessary	Wheeled pusher

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EXP60K3 Fast DC Wallbox Charger

60kW DC Charging Power with Two DC Charging Connectors + (o) 22kW AC Type 2 Charging Cable

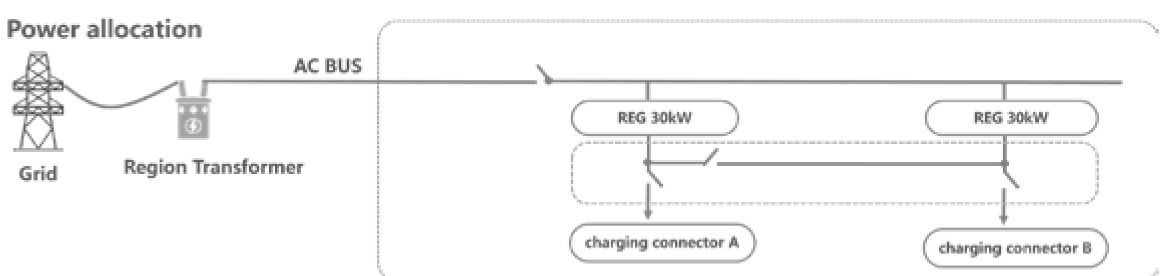
Introduction

EXP60K3 DC charger is specially designed with wall mounted and pedestal installations and fully compatible with CCS/NACS/CHAdeMO/GBT EV charging standard. It features small size, light weight and easy to install and move, user friendly interface, IP55 protection, robust and durable design for various outdoor applications. It can deliver up to 60kW charging power and 150V to 1000V charging voltage. It can be engineered with optional 22kW/32A AC Type 2 charging cable.

Main Features

- CE/RED, CB, TUV US/CA, Energy Star and OCA certification
- WEB/APP Maintenance Backend System enhances the operation and maintenance effectiveness
- CCS1, CCS2, GBT, CHAdeMO and NACS support
- Wall mounted, pedestal installation
- Optional cable retractor, credit card (POS), VIN code support
- IP55 protection, reliable & robust
- High power density with best space efficiency
- Three vehicles charging at the same time
- AC Type 2 charging cable configuration support
- BABA support

Application



Technical Parameters

Input Parameters

AC frequency	45-65Hz
AC voltage	CE: 3 phase+N+PE, 380/400Vac±10% UL: 3 phase+PE, 480Vac±10%
Power factor	>0.99
Total harmonic current	<3% (rated input)

Output Parameters

Output DC voltage range	DC 150V-1000V
Max DC output current	200A
Max DC output power	60kW
Constant power voltage range	300~1000V
Max system efficiency	>95% (rated input and output)
Optional AC charging output	AC charging cable 32A/3P/22kW

Working Environment

Ambient temperature	-30~+70°C, full power output below 50°C, power derating 5%/°C above 50°C
Storage temperature	-40~+75°C
Working humidity	0~95%
Working altitude	2000 m

Standard

Backend	OCPP 1.6J, support firmware update to 2.0x, OTA support, P&C/smart charge/power management support
CCS PLC communication	DIN70121 and ISOI5118
CHAdeMO	CHAdeMO V1.2
GBT	GB/T 18487.1-2015, GB/T 27930-2015, GB/T 20234.1~3-2015
Certification	TUV US+CA, Energy Star, TUV CE, RED, CB, OCA UL 2202/CSA C22.2; EN IEC61851-21-2 Class B; EN IEC61851-1/EN IEC61851-23/EN 61851-24
Protection class	IP55/IK10

Function and Interface

HMI	Tempered glass protective 7" TFT touch screen LCD, RGB LED, EPO
Metering	DC energy meter, DL/T645 standard, optional MID meter
Monitor interface	LAN and LTE wireless modem support
Charging mode	Auto basis, time basis, amount basis, energy basis, SOC basis
Power switch mode	Two DC charging connectors charging, AC Type 2 cable charging in parallel
Payment	RFID, Mobile APP, POS (opt)
MBE function support	WEB side Maintenance Backend System enhances the operation and maintenance effectiveness

Mechanical

Dimensions (W*H*D mm)	705 * 1100 * 240 mm
Weight (kg)	92 kg (excluding power module, power module is 15.5kg)

Charging Protection

- Overcurrent protection, short circuit protection, overvoltage protection, undervoltage protection
- Insulation monitoring, protect earth connect monitoring
- Reverse polarity protection, overtemperature protection
- RCDB for AC charging cable configuration protection

Configuration

Charger module configuration	REG1K0100G2 30kW1000V charger power module
Charging connector configuration	One or two CCS/NACS/GBT connectors configuration, CHAdeMO + CCS/NACS/GBT connectors configuration, optional AC Type 2 charging cable
Charging connector capacity	GBT: 200A CCS: 200A NACS: 200A CHAdeMO: 125A AC: 32A/22kW
Optional pedestal	Pedestal with two cable retractors and the light bar

EXP150K3-HD IP65 Fast DC Charger

150kW DC Charging Power with Two DC Charging Connectors + (o) 22kW AC Charging Socket

Introduction

EXP150K3-HD DC charger is all in one design and fully compatible with CCS2/CHAdeMO/GBT EV charging standard. User friendly interface, Heat exchange technical to get IP65 protection for various harsh environment applications, robust and durable design for various outdoor applications. It can deliver up to 150kW charging power and 150V to 1000V charging voltage. It also can be engineered with optional 22kW/32A AC Type 2 charging socket.

Main Features

- IP65 protection, reliable & robust, salt fog resistant, dust prevention, waterproof
- AC Type 2 charging socket configuration support
- Optional cable retractor, credit card (POS), VIN code support
- WEB/APP Maintenance Backend System enhances the operation and maintenance effectiveness
- CCS2, GBT and CHAdeMO support
- Three vehicles charging at the same time
- CE, RED, CB, UKCA and OCA certification

Technical Parameters

Input Parameters

AC frequency	45-65Hz
AC voltage	3-phases+N+PE, 380/400Vac±10%
Power factor	>0.99
Total harmonic current	<3% (rated input)

Output Parameters

Output DC voltage range	150V-1000V
Max DC output current and power	200A, 150kW
Optional AC charging output	AC type 2 charging socket 32A/3P/22kW
Constant power voltage range	300~1000V
Max system efficiency	>95% (rated input and output)
Noise	<65dB @rated input and output, 30°C, full power

Working Environment

Ambient temperature	-30~+65°C, full power output below 45°C, power derating 5%/°C above 45°C
Storage temperature	-40~+75°C
Working humidity	0~95%
Working altitude	2000m

Standard

Backend	OCPP 1.6J, support firmware update to 2.0x, OTA support, P&C/ Smart charge/ Power management support
CCS PLC communication	DIN70121 and ISO15118
CHAdeMO	CHAdeMO V1.2
GBT	GB/T 18487.1-2015, GB/T 27930-2015, GB/T 20234.1~3-2015
Certification	TUV CE, RED, CB, UKCA, OCA
	EN61851-21-2 Class A; EN 61851-1/EN 61851-23/EN 61851-24
Protection class	IP65/IK10 (touch screen IK08)

Function and Interface

HMI	Tempered glass protective 7" TFT touch screen LCD, three RGB LED, EPO
Metering	DC energy meter, GB/T 29318 and JJG1149 standard, optional MID meter
Monitor interface	LAN and LTE wireless modem support
Charging mode	Auto basis, time basis, amount basis, energy basis, SOC basis
Power switch mode	Intelligent power switch matrix between two DC charging connectors
Payment	RFID, Mobile APP, POS (opt)
MBE function support	WEB side Maintenance Backend System enhances the operation and maintenance effectiveness

Mechanical

Dimensions (W*H*D mm)	700 * 1750 * 760mm
Weight (kg)	280kg

Charging Protection

Overcurrent protection, short circuit protection, overvoltage protection, undervoltage protection
Insulation monitoring, protect earth connect monitoring
Reverse polarity protection, overtemperature protection
RCD-A main protection and RCD-B for AC connector protection

Configuration

Charger module configuration	5 * REG1K0100G2 30kW1000V charger power module
Charging connector configuration	2 * CCS2/GBT or CCS2+GBT/CHAdeMO, optional AC Type 2 charging socket
Charging connector capacity	GBT: 250A CCS2: 200A CHAdeMO: 125A/200A AC: 32A/22kW
Optional cable management	Top assembly cable retractors and light bar

EXP180K2-FD/EXP240K5-FD Fast DC Charger

180kW and 240kW DC Charging Power with Two DC Charging Connectors + (o)22kW AC Charging Socket

Introduction

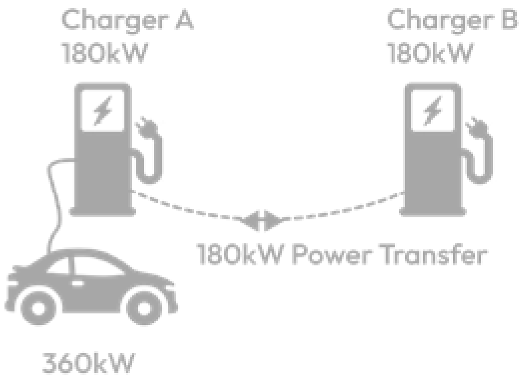
EXP180K2-FD and EXP240K5-FD DC charger is all in one design and fully compatible with CCS/NACS/CHAdeMO/GBT EV charging standard. It features small size, light weight and easy to install and move, user friendly interface, IP55 protection, robust and durable design for various outdoor applications. It can deliver up to 180kW and 240kW charging power and 150V to 1000V charging voltage. It can be engineered with optional 22kW/32A AC Type 2 charging socket and Eichrecht certification configuration.

Main Features

- CE, RED, CB and TUV US/CA certification, Energy Star certification, OCA certification
- WEB/APP Maintenance Backend System enhances the operation and maintenance effectiveness
- CCS1, CCS2, GBT, CHAdeMO and NACS support
- CoolRing™ power sharing between two chargers[1]
- Optional cable retractor, credit card (POS), VIN code support
- Three vehicles charging at the same time[1]
- AC Type 2 charging socket configuration support[1]
- IP55 protection, reliable & robust

Note 1: only EXP180K2 support.

CoolRing™ work with two 180kW chargers



Each charging connector can get full 360kW power when CoolRing works

Charger A-DC1	ChargerA-DC1	Charger B-DC1	Charger B-DC2
90kW	90kW	90kW	90kW
180kW	0	180kW	0
180kW	90kW	90kW	0
180kW	180kW	0	0
360kW	0	0	0

Technical Parameters

Input Parameters

AC frequency	45-65Hz		
AC voltage	3-phases+(N)+PE, 380/400Vac±10% (CE), 480Vac±10% (NA)		
Power factor	>0.99		
Total harmonic current	<3% (rated input)		

Output Parameters

Output Parameters	EXP180K2-FD	EXP180K2-FT	EXP240K5-FD
Output DC voltage range	DC 150V-1000V		
Max DC output current	600A@300V/360A@500V/180A@1000V		600A@400V/480A@500V/240A@1000V
Max DC output power	180kW		240kW
Constant power voltage range	300~1000V		300~1000V
Max system efficiency	>95% (rated input and output)		>95.5% (rated input and output)
AC connector		AC Type 2 32A/22kW/3P	

Working Environment

Ambient temperature	-30°C ~ +70°C, full power output below 50°C, power derating 5%/°C above 50°C
Storage temperature	-40°C ~ +75°C
Working humidity	0-95%
Working altitude	2000m

Standard

Backend	OCPP 1.6J, support firmware update to 2.0x, OTA support, P&C/smart charge/power management support
CCS PLC communication	DIN70121 and ISO15118
CHAdeMO	CHAdeMO V1.2
GBT	GB/T 18487.1-2015, GB/T 27930-2015, GB/T 20234.1~3-2015
Certification	TUV US+CA, Energy Star, TUV CE, RED, CB, OCA , NMI EICHRECHT
	UL2202, EN61851-21-2 Class A; EN 61851-1/EN 61851-23/EN 61851-24
Protection class	IP55/IK10 (touch screen IK08)

Function and Interface

HMI	Tempered glass protective 7" TFT touch screen LCD, three RGB LED, EPO
Metering	DC energy meter, GB/T 29318 and JJG1149 standard, optional MID and NMI certificatiion meter
Monitor interface	LAN and LTE wireless modem support
Optional CoolRing™ Power transfer	Realize the power transfer between two chargers, each charging connector can get the total charging power
Power switch mode	Intelligent power switch matrix between two DC charging connectors
Payment	RFID, Mobile APP, POS (opt)
MBE function support	WEB side Maintenance Backend System enhances the operation and maintenance effectiveness

Mechanical

Dimensions (W * H * D mm)	700 × 1750 ×750 mm
Weight (kg)	260 kg

Charging Protection

- Overcurrent protection, short circuit protection, overvoltage protection, undervoltage protection
- Insulation monitoring, Protect earth connect monitoring
- Reverse polarity protection, overtemperature protection
- RCD-A main protection and RCD-B for AC connector protection

Configuration

Configuration	Total 6 power module sockets configuration	
Charger module configuration	REG1K0100G/U 30kW1000V charger power module	REG1K0110U 40kW1000V charger power module
Charging connector configuration	2 * CCS/NACS/GBT or CCS/NACS+GBT/CHAdeMO, optional AC type 2 charging socket	
Charging connector capacity	GBT: 250A/400A CCS2: 250A/375A CCS1: 300A NACS: 200A/300A CHAdeMO: 125A/200A AC: 32A/22kW	
Optional cable management	Top assemble cable retractors and the light bar	

EXP180K4-LD IP65 Fully Liquid Cooled Fast DC Charger

180kW DC charging power with two DC charging connectors

Introduction

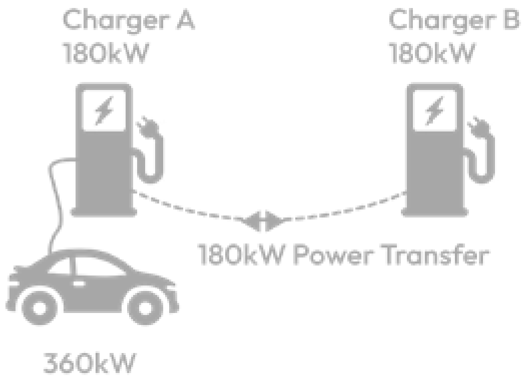
EXP180K4-LD DC charger is all in one design and fully compatible with CCS2/CHAdeMO/GBT EV charging standard. It features full liquid cooled design with low audio noise, high protection, high reliability and highly integrated, it employs reliable liquid-cooled power modules and reaches IP65 protection rating for various harsh environment applications. It can deliver up to 180kW charging power and up to 500A charging current with the output voltage from 150V to 1000Vdc.

Main Features

- IP65 protection, reliable & robust, salt fog resistant, dust prevention, waterproof
- WEB/APP Maintenance Backend System enhances the operation and maintenance effectiveness
- CCS2, GBT and CHAdeMO support
- Optional one 500A liquid cooling connector
- Optional cable retractor, credit card (POS), VIN code support
- 60dB ultra low noise
- CE, RED, CB and OCA certification

CoolRing™ work with two 180kW chargers

Each charging connector can get full 360kW power when CoolRing works



Each charging connector can get full 360kW power when CoolRing works

Charger A-DC1	ChargerA-DC1	Charger B-DC1	Charger B-DC2
90kW	90kW	90kW	90kW
180kW	0	180kW	0
180kW	90kW	90kW	0
180kW	180kW	0	0
360kW	0	0	0

Technical Parameters

Input Parameters

AC frequency	45-65Hz
AC voltage	3-phases+N+PE, 380/400Vac±10%
Power factor	>0.99
Total harmonic current	<3% (rated input)

Output Parameters

Output DC voltage range	DC 150V-1000V
Max DC output current	600A@300V/360A@500V/180A@1000V
Max DC output power	180kW
Constant power voltage range	300~1000V
Max system efficiency	>95% (rated input and output)

Working Environment

Ambient temperature	-30°C ~ +65°C, full power output below 50°C, power derating 5%/°C above 50°C
Storage temperature	-40°C ~ +75°C
Working humidity	0-95%
Working altitude	2000m

Standard

Backend	OCPP 1.6J, support firmware update to 2.0x, OTA support, P&C/smart charge/power management support
CCS PLC communication	DIN70121 and ISO15118
CHAdeMO	CHAdeMO V1.2
GBT	GB/T 18487.1-2015, GB/T 27930-2015, GB/T 20234.1~3-2015
Certification	TUV CE, RED, CB, OCA EN61851-21-2 Class A; EN 61851-1/EN 61851-23/EN 61851-24
Protection class	IP65/IK10 (touch screen IK08)

Function and Interface

HMI	Tempered glass protective 7" TFT touch screen LCD, three RGB LED, EPO
Metering	DC energy meter, GB/T 29318 and JJG1149 standard, optional MID meter
Monitor interface	LAN and LTE wireless modem support
Optional CoolRing™ power transfer	Realize the power transfer between two chargers, each charging connector can get the total charging power
Power switch mode	Intelligent power switch matrix between two DC charging connectors
Payment	RFID, Mobile APP, POS (opt)
MBE function support	WEB side Maintenance Backend System enhances the operation and maintenance effectiveness

Mechanical

Dimensions (W * H * D mm)	700 × 1995 × 850 mm
Weight (kg)	528 kg

Charging Protection

Overcurrent protection, short circuit protection, overvoltage protection, undervoltage protection

Insulation monitoring, protect earth connect monitoring

Reverse polarity protection, overtemperature protection

RCD-A main protection

Configuration

Charger module configuration	LRG1K0100G 30kW1000V liquid cooled charger power module, total 6 power module sockets
Charging connector configuration	2 * CCS2/GBT or CCS2+GBT/CHAdeMO
Charging connector capacity	GBT: 250A/400A/500A CCS2: 250A/375A/500A CHAdeMO: 125A/200A

EXP600K3 Air Cooled Split Type High Power Charger

600kW Power Cube/12 Charging Connectors, Overhead Charging Dispenser Solution

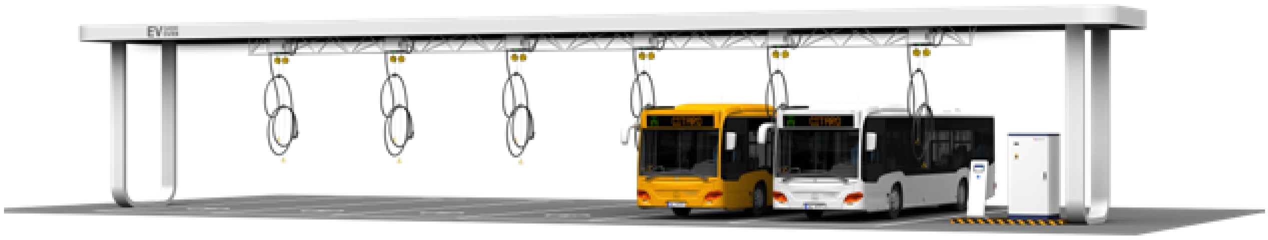
Introduction

EXP600K3-FWR5 Split type high power charger is specially designed for the EV charger operators with split and multi charging point applications. It supports CCS/NACS/CHAdeMO/GBT compatible charging connectors, 600kW high power output and max 12 charging connectors with 6 dual-connectors dispensers or 12 single-connector dispensers. It also supports 500A liquid cooled charging connector with full power. Intelligent power switching matrix can realize the power modules power switch between the charging connectors to maximize the power using efficiency. The overhead dispenser for the bus station solution is also available with the same power cube.

Main Features

- Building-block modular design, simplified & standardized interface
- Max 12 charging channels configuration, 300A or 500A configuration
- TUV US/CA, CE, RED, CB and OCA certification, Energy Star certification
- Power cube IP54, stand floor dispenser IP55, overhead dispenser IP65, reliable & robust
- Patented U-Type air duct design
- Customizable dispenser, overhead dispenser for bus station solution
- WEB/APP Maintenance Backend System enhances the operation and maintenance effectiveness
- High power density with best space efficiency
- Power cube 65dB, stand floor dispenser 60dB, overhead dispenser 55dB
- BABA support

Overhead Micro Charging Dispenser Solution for the Bus Station



Technical Parameters

Power Cube

		EXP600K3-FWR5
AC input	AC input frequency/voltage	45-65Hz/ 3-phase+(N)+PE /260Vac-530Vac
	AC max input current	CE: 2*480A@380Vac/UL:2*380A@480Vac
	Power factor	>0.99
	Total harmonic current	<5% (rated input)
DC output	Output DC voltage range	150V-1000V
	Max DC output power	600kW@300V~1000V
	Max DC output current	2000A@300V, 1200A@500V, 600A@1000V
	Max system efficiency	>95.0% (rated input and output)
	Output channel	Max 12 output charging channels as the input of the charging dispensers, 10 output charging channels (opt)
	Dispenser support	Max 6 dual-connectors dispensers or 12 single-connector dispensers
HMI	RGB LED, E-STOP	
Mechanical	Dimensions: W * H * D mm = 1050 * 2200 * 1150 mm Weight: 690kg + 20 * 15.5kg = 1000 kg (full power modules)	
Protection class	IP54	
Noise	<60dB (@25°C, rated input and output), max noise < 65dB	
Power switch	Power can switch among the power module groups to each output channel to maximize the power efficiency	
Power module	Max 20 module slots	REG1K0100G2 30kW/1000V power module
HMI kiosk	Optional, tempered glass protective 10" capacitive touch screen LCD, RFID, POS (opt)	

Charging Dispenser

		Stand Floor Dispenser				Overhead Dispenser
		EXP200/250 FDC	EXP300-FDC	EXP400-FDC	EXP500-LDC/LLDC	EXP200/250/300-FSW
DC input	DC input voltage range	150V~1000V				
	DC input channel	2 channel DC bus	2 channel DC bus	2 channel DC bus	2 channel DC bus	1 channel DC bus
DC output	Output charging cable	2	2	2	2	1
	Output DC voltage range	150V~1000V				
	Max DC output current	2 * 200A (opt 250A)	2*300A	2*375A	500A+200A/500A +300A/opt 500A	200A (opt 250A/300A)
HMI	Tempered glass protective 7" capacitive touch screen LCD for stand floor dispenser, RGB panel LED, E-STOP, RFID, POS (opt)					
Metering	DC energy meter, opt NMI Eichrecht certification configuration					
Monitor	OCPP 1.6J, support update to 2.0, P&C, smart charge and power management support, LAN 100M and LTE wireless modem support					
Cable management	Top assembly two cable retractors and light bar for stand floor dispenser. Select one or two cable retractors based on cable length for overhead dispenser					
MBE function support		WEB side Maintenance Backend System enhances the operation and maintenance effectiveness				
Charging protection		Overcurrent/short circuit/overvoltage/undervoltage/RCD/insulation/overtemperature protection				
		Reverse polarity protection				
Mechanical	Dimensions: W*H*D mm	500*1800*260/195kg/IP55/IK10				763*415*240/50kg/IP65
Charging connector configuration		2*GBT/CCS/NACS or CCS+NACS/GBT/CHAdeMO or 1*GBT/CCS/NACS/CHAdeMO				CCS or GBT or NACS
Charging connector capacity		GBT:250A (opt 400/600A) , CCS:200A (opt 250/300/375/500A), NACS:200A (opt 300A/380A) , CHA:125A (opt 200A)				
Noise		0			60dB	0

Standard

Charging standard	GBT 27930-2011 and 27930-2015, CHAdeMO V1.2/V2.0, DIN70121 and ISO15118, OCA
EMC/safety	UL2202, CSA C22.2; CE/RED/CB; EN 61851-21-2/EN 61000-6-4/EN 61000-6-2 Class A; EN 61851-1/EN 61851-23/EN 61851-24

Working Environment

Ambient temperature	Working temperature range -30~+70°C, full load temperature range -25~+50°C, power derating 5%/°C above 50°C
Storage temperature	-40~+75°C
Working humidity	0 ~ 95%
Working altitude	2000 m

EXP600K4 Liquid Cooled Split Type High Power Charger

600kW Liquid Cooled Power Cube/12 Charging Connectors, Overhead Charging Dispenser Solution

Introduction

EXP600K4-LWR5 split type high power charger is specially designed for the EV charger operator with split and multi charging point application. CCS2/ CHAdeMO/ GBT standard, 600kW high power output and max 12 charging connectors to support 6 dual-connectors dispensers or 12 single-connector dispensers. Optional 500A liquid cooled charging connector and each charging connector can get max power through the intelligent power switching between power module groups to maximize the power efficiency. Full liquid cooled power module application for extreme high protection and high reliability . The overhead charging dispenser for the bus station solution is also available with the same power cube.

Main Features

- Building-block modular design, simplified & standardized interface
- IP65 protection, reliable & robust, salt fog resistant, dust prevention, waterproof
- CE, RED, CB and OCA certification, Optional NMI Eichrecht certification configuration
- Patented U-Type air duct design
- Power cube 65dB, stand floor dispenser 60dB, overhead dispenser 55dB
- Customizable dispenser, overhead dispenser for bus station solution
- Max 12 charging channels configuration, 300A or 500A configuration
- WEB/APP Maintenance Backend System enhances the operation and maintenance effectiveness
- Fully liquid cooled

Technical Parameters

Power Cube

		EXP600K4-LWR5
AC input	AC input frequency/voltage	45-65Hz/ 3-phase+(N)+PE / 260Vac-530Vac
	AC max input current	2*480A@380Vac
	Power factor	>0.99
	Total harmonic current	<5% (rated input)
DC output	Output DC voltage range	150V-1000V
	Max DC output power	600kW@300V~1000V
	Max DC output current	2000A@300V, 1200A@500V, 600A@1000V
	Max system efficiency	>95.5% (rated input and output)
	Output channel	Max 12 charging channels as the input of the charging dispensers, 10 output charging channels (opt)
	Dispenser support	Max 6 dual-connectors dispensers or 12 single-connector dispensers
HMI	RGB LED, E-STOP	
Mechanical	Dimensions: W * H * D mm = 2000 * 2200 * 850 mm Weight: 1200kg + 20 * 30kg = 1800 kg (full power modules and full liquid)	
Protection class	IP65 (excluding heat exchange chamber)	
Noise	<60dB (@25°C, rated input and output), max noise <65dB	
Power switch	Intelligent power switch among the power module groups, can switch to each output channel to maximize the power efficiency	
Power module	Max 20 module slots	LRG1K0100G 30kW/1000V liquid cooled power module

Charging Dispenser

Charging Dispenser		Stand Floor Dispenser				Overhead Dispenser
		EXP200/250 FDC	EXP300-FDC	EXP400-FDC	EXP500-LDC/LLDC	EXP200/250/300-FSW
DC input	DC input voltage range	150V~1000V				
	DC input channel	2 channel DC bus	2 channel DC bus	2 channel DC bus	2 channel DC bus	1 channel DC bus
DC output	Output charging cable	2	2	2	2	1
	Output DC voltage range	150V~1000V				
	Max DC output current	2 * 200A (opt 250A)	2*300A	2*375A	500A+200A/500A +300A/opt 500A	200A (opt 250A/300A)
HMI	Tempered glass protective 7" capacitive touch screen LCD for stand floor dispenser, RGB panel LED, E-STOP, RFID, POS (opt)					
Metering	DC energy meter, opt NMI Eichrecht certification configuration					
Monitor	OCPP 1.6J, support update to 2.0, P&C, smart charge and power management support, LAN 100M and LTE wireless modem support					
Cable management	Top assembly two cable retractors and light bar for stand floor dispenser. Select one or two cable retractors based on cable length for overhead dispenser					
MBE function support		WEB side Maintenance Backend System enhances the operation and maintenance effectiveness				
Charging protection		Overcurrent/short circuit/overvoltage/undervoltage/RCD/insulation/overtemperature protection Reverse polarity protection				
Mechanical	Dimensions: W*H*D mm	500*1800*260/195kg/IP55/IK10				763*415*240/50kg/IP65
Charging connector configuration		2*GBT/CCS/NACS or CCS+NACS/GBT/CHAdeMO or 1*GBT/CCS/NACS/CHAdeMO				CCS or GBT or NACS
Charging connector capacity		GBT:250A (opt 400/600A) , CCS:200A (opt 250/300/375/500A), NACS:200A (opt 300A/380A) , CHA:125A (opt 200A)				
Noise		0			60dB	0

Standard

Charging standard	GBT 27930-2011 and 27930-2015, CHAdeMO V1.2/V2.0, DIN70121 and ISO15118, OCA
EMC/safety	CE/RED/CB; EN 61851-21-2/EN 61000-6-4/EN 61000-6-2 Class A; EN 61851-1/EN 61851-23/EN 61851-24

Working Environment

Ambient temperature	Working temperature range -30~+70°C, full load temperature range -25~+50°C, power derating 5%/°C above 50°C
Storage temperature	-40~+75°C
Working humidity	0 ~ 95%
Working altitude	2000 m

Overhead Charging Dispenser System

200kW/250kW/300kW/350kW/375kW/380kW Output Power

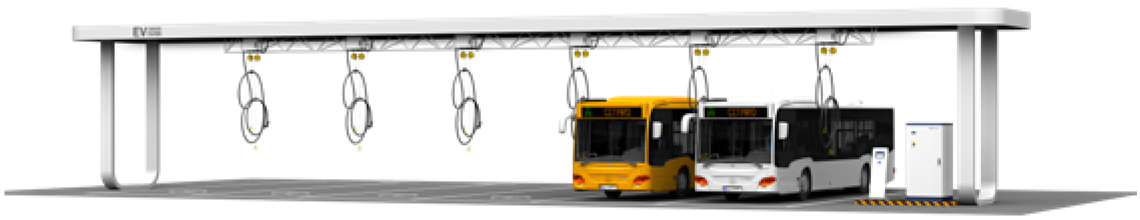
Introduction

The Overhead Charging Dispenser system is specially designed for the application of split and multi electric vehicle charging points by operators, and adopts CCS and NACS standard. It can support configurations of up to 200A (CCS2, CCS1, NACS)/250A(CCS2)/300A(CCS2, CCS1, NACS)/350A(CCS1, NACS)/375(CCS2)/380A(NACS) air-cooled charging connectors. The maximum output power corresponds to 200kW, 250kW, 300kW, 350kW, 375kW and 380kW respectively.

Main Features

- Full safety functions with output contactor and fuse, ESD, SPD, leakage switch, insulation detector, software logic for multiple protection
- Easily configure the output power up to 380kW and the output voltage up to 1000V
- Fast charge all electric vehicles compliant with CCS/NACS standards
- IP65, high protection and high reliability, -30°C~+50°C full power charging
- User friendly, coupling the charger's output plug in the EV for automatic starting
- WEB/APP Maintenance Backend System enhances the operation and maintenance effectiveness
- LAN and CAN wireless support, Mobile App payment support
- Battery charging state displayed on the HMI

Application Scenario



Technical Parameters

		EXP400-FSW	EXP300-FSW	EXP250-FSW	EXP200-FSW
Nominal input	Phases/lines	(DC+, DC-) +PE	(DC+, DC-)+PE	(DC+, DC-)+PE	(DC+, DC-) +PE
	Voltage range	Max 1000Vdc	Max 1000Vdc	Max 1000Vdc	Max 1000Vdc
	Current	350A/375A/380A	300A	250A	200A
	Power	350kW/375A/380kW	300kW	250kW	200kW
	System efficiency	≥ 99% (full load)	≥ 99% (full load)	≥ 99% (full load)	≥ 99% (full load)
DC output CCS2	Connector	CCS1, CCS2, NACS	CCS1, CCS2, NACS	CCS2	CCS1, CCS2, NACS
	Voltage	150~1000Vdc	150~1000Vdc	150~1000Vdc	150~1000Vdc
	Current	CCS1 350A CCS2 375A NACS 350A,380A	CCS1 300A CCS2 300A NACS 300A	CCS2 250A	CCS1 200A CCS2 200A NACS 200A
	Nominal power	350kW/375kW/380kW	300kW	250kW	200kW
Auxiliary power input	Voltage	UL:480VAC; CE:230VAC	UL:480VAC; CE:230VAC	CE:230VAC	UL:480VAC; CE:230VAC
	Current	UL:0.6A CE:1A	UL:0.6A CE:1A	CE:1A	UL:0.6A CE:1A
Cabinet	Dimensions (W*D*H)	763*240*415 (mm)	763*240*415 (mm)	763*240*415 (mm)	763*240*415 (mm)
	Weight	50kg	48kg	46kg	44kg
	Protection degree	NEMA 4X/IP65	NEMA 4X/IP65	IP65	NEMA 4X/IP65
Command unit	Local interface	Status LED	Status LED	Status LED	Status LED
	Communication	CAN/Ethernet	CAN/Ethernet	CAN/Ethernet	CAN/Ethernet
Environment conditions	Operating temperature[1]	-30°C~+50°C	-30°C~+50°C	-30°C~+50°C	30°C~+50°C
	Transportation/storage temperature	-40°C~+70°C	-40°C~+70°C	-40°C~+70°C	-40°C~+70°C
	humidity	5%RH~95%RH	5%RH~95%RH	5%RH~95%RH	5%RH~95%RH
	Place of installation	Indoor/outdoor	Indoor/outdoor	Indoor/outdoor	Indoor/outdoor
	Altitude	2000m	2000m	2000m	2000m
	Sound noise	≤55dB (nominal input/ output power, the environment temperature is 25°C.)	≤55dB (nominal input/ output power, the environment temperature is 25°C.)	≤55dB (nominal input/ output power, the environment temperature is 25°C.)	≤55dB (nominal input/ output power, the environment temperature is 25°C.
	Atmospheric pressure	80KPa~110KPa	80KPa~110KPa	80KPa~110KPa	80KPa~110KPa
	Overvoltage category	III	III	III	III
	Protection class	Class I	Class I	Class I	Class I

Note 1: The Charging Dispenser provides full output power up to 50°C, output power derating 5%/°C above 50°C.

Overhead Charging Dispenser System Model

Model	Configuration	Certification
EXP200-FSW-E	CCS2/200A/200kW	CE/TUV, CB
EXP250-FSW-E	CCS2/250A/250kW	CE/TUV, CB
EXP300-FSW-E	CCS2/300A/300kW	CE/TUV, CB
EXP400-FSW-E	CCS2/375A/375kW	CE/TUV, CB
EXP200-FSW-U2	CCS1/200A/200kW	cTUVus
EXP300-FSW-U2	CCS1/300A/300kW	cTUVus
EXP400-FSW-U2	CCS1/350A/350kW	cTUVus
EXP200-FSW-N2	NACS/200A/200kW	cTUVus
EXP300-FSW-N2	NACS/300A/300kW	cTUVus
EXP400-FSW-N2	NACS/380A/380kW; NACS/350A/350kW	cTUVus

E-mail: info@unicom-bass-systems.com
unciom-bass-systems.com

IHS540K2 Battery Energy Storage Charging Power Cube

540kW Charging Power and 6 Charging Connectors

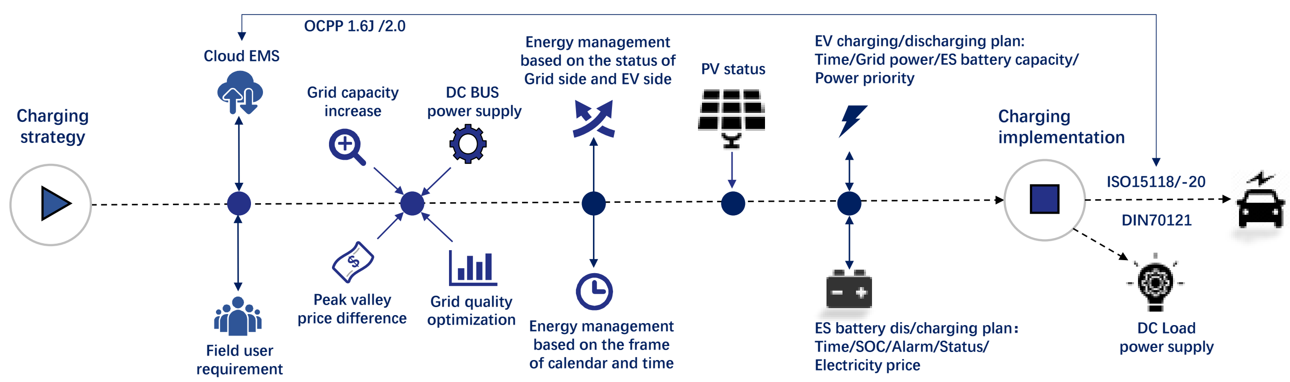
Introduction

The battery energy storage charging power cube is the key block of the energy storage charging solution and supports max 540kW EV charging power capacity with max 18 power modules. It features totally electrical isolation design between the grid, battery, PV and EV charging for maximum safety assurance and a mixed AC and DC bus design between the AC grid and the battery to ensure optimal power efficiency, system flexibility and safety.

Main Features

- Building-block design, easy installation and maintenance
- Independent battery group access with separate battery bus isolation
- Easy new energy access and optional DC bus power supply
- WEB/APP Maintenance Backend System enhances the operation and maintenance effectiveness
- All in one design, improving space efficiency and cost efficiency
- TUV CE and TUV US/CA certification

Energy Management Strategy



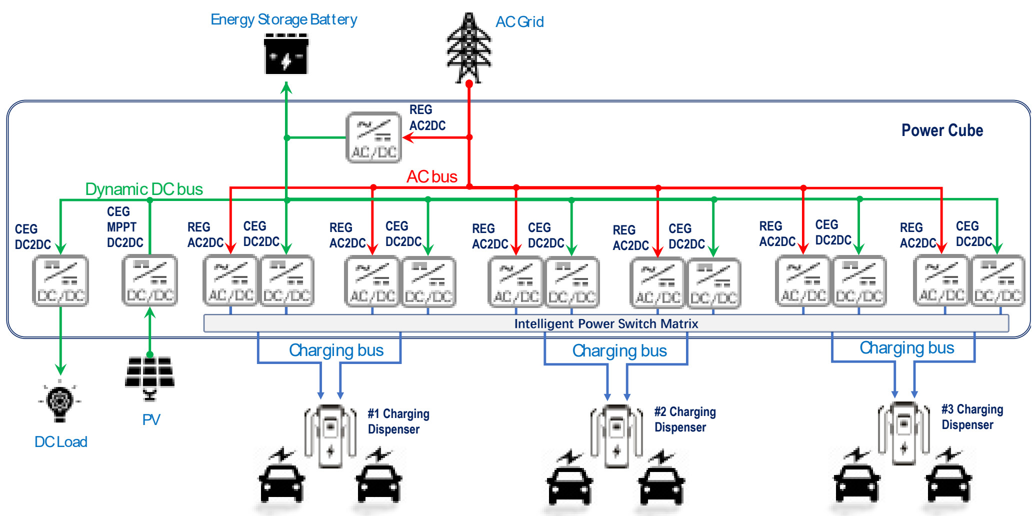
Technical Parameters

IHS540K2 540kW Power Cube

AC grid access	AC input voltage	45-65Hz/3-phase+(N)+PE/260Vac-530Vac
	AC max input power	Max 360kW
	AC distribution	AC grid charging power to energy storage battery is max 120kW, to EV is max 360kW
Energy storage battery access	Battery group access	1 group
	Battery capacity/group	200kWh/280Ah/627V~806V/0.5C dis/charge/max 0.75C discharge 30 min
	Battery charging power from AC grid	Max 4 * 30kW =120kW (4 *30kW ACDC power module configuration)
	Battery discharging channel to EV	Max 6 channels
	Battery B2V EV charging power	Max 6 * 30kW =180kW (6 *30kW DCDC power module configuration)
Electric vehicle charging	EV charging point	Max 6 charging connectors, 3 dispensers with dual charging connectors
	EV charging power	Max 360 kW (180kW from AC grid, 180kW from ES battery)
	EV charging power transfer	Power switch among the power module groups to maximize the power efficiency
	EV charging voltage	150V~1000V
PV access	Access channel	2 channel
	Access power	Max 2 *30kW MPPT
Metering	AC grid side	1 AC energy meter
HMI	7 " TFT touch screen, 5 LED, E-STOP	
Dimensions	W * H * D mm = 1000 * 2200 * 1150 mm	Weight: 550 + 18*22.5kg = 955kg (full power modules)
DC power supply	Optional function, support 150V~1000V DC load (max 60kW)	

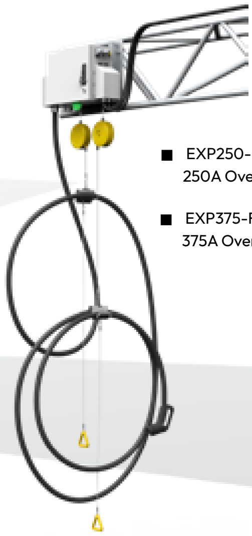
Power Module Configuration

Max 18 power module slots for the configuration	CEG1K0100G-MPPT	DC2DC converter, 30kW/1kV, MPPT support	Max 2*30kWp solar power input support
	CEG1K0100G	DC2DC converter, 30kW/1kV	Max 6*30kW ES battery discharging/EV charging configuration and max 2*30kW DC load support
	REG1K0100G	AC2DC rectifier, 30kW/1kV	Max 8*30kW EV charging configuration and max 4*30kW ES battery charging configuration



Liquid Cooled Energy Storage Charging Solution

400kW Charging Power/6 Charging Connectors/350kWh Battery Energy Storage Charger

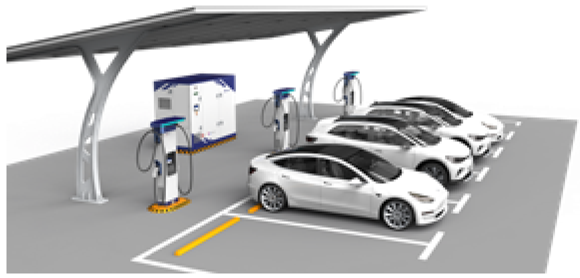


- EXP250-FSW-E
  250A Overhead Charging Dispenser
- EXP375-FSW-E
  375A Overhead Charging Dispenser

- IHEBS470K7G-350LL3C
  470kW/350kWh Energy Cube
- EXP250-FDC 250A Charging Dispenser
- EXP375-FDC 375A Charging Dispenser

Introduction

The energy storage charging system employs LFP battery for energy storage and through the local and cloud EMS, it helps balance the power supply and demand between the grid, battery, EV charging and renewable energy access. Fully liquid cooled solution for battery groups can realize the 1P discharge performance to meet the limit AC grid power but high power charging ability with cost efficiency and space saving. It carries significant values in grid peak load shaving, renewable energy resource access and grid power capacity expansion with battery discharge power to grid.



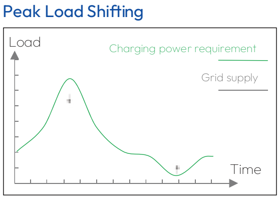
Main Features

- Full liquid cooled design, from power module to the battery pack and charging cable
- Highly integrated all in one design, PV, ESS, EV charging and V2G functions
- Friendly noise management
- Micro-grid with high charging power
- Full electric safety design with the total electric isolation between the grid, battery, EV and solar power input
- Easy site-to-site transfer
- 1P/1C fast charge and discharge

Solution Values

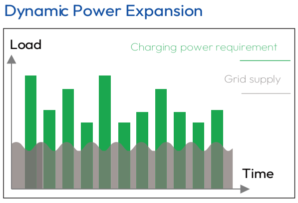
- Grid peak load shaving
- Grid capacity expansion
- Backup power supply by B2G
- New energy convenient access
- HPC with limit grid power
- Off grid still keep charging

Peak Load Shifting



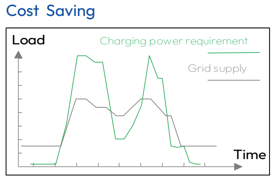
Charge in off-peak hours and discharge in peak hours, achieving peak-valley arbitrage and saving electricity costs.

Dynamic Power Expansion



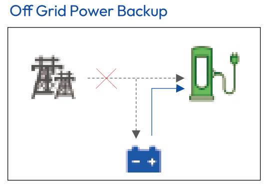
For intermittent high-power loads, the battery system can support the dynamic power expansion and stable grid power requirement.

Cost Saving

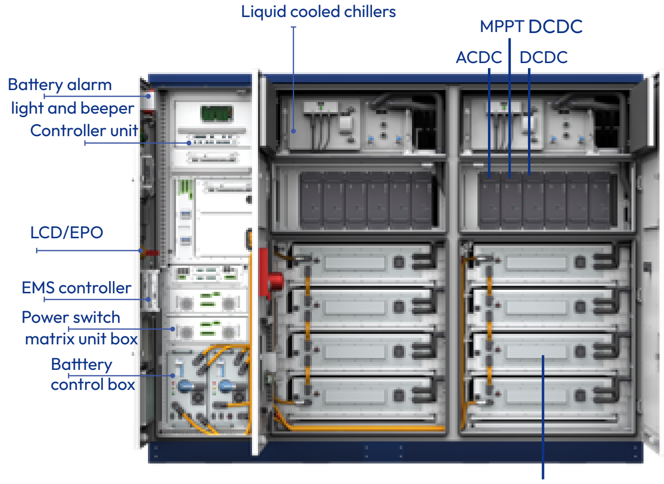


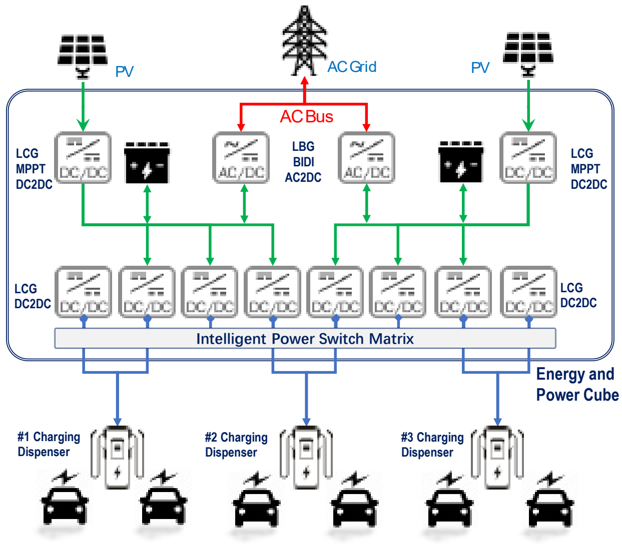
Participate in demand side management and obtain government subsidies; reduce peak power and save basic electricity bills.

Off Grid Power Backup



When the grid powers off, the battery system can still keep power supply for EV charging.





Technical Parameters

IHEBS470K7G-350LL3C 470kW/350kWh BES All in One Battery Charger

AC grid access	AC input voltage	45-65Hz/3-phase+(N)+PE/260Vac-530Vac
	AC max input power	60kW, optional 70kW bidirectional
	AC max output power (optional)	70kW bidirectional power
Electric vehicle charging	EV charging point	Max 6 connectors, 3 dispensers with dual-connectors or 6 single connector dispensers
	EV charging power	Max 400 kW
	EV charging power transfer	Power switch between the power module groups to maximize the power efficiency
	EV charging voltage	150V~1000V
PV access	Access channel	Max 2 channels
	Access power	Max 2 * 40kW MPPT
Metering	AC grid side	1 AC energy meter, optional MID meter
HMI	10" TFT touch screen, 2 RGB LED, E-STOP, batt alarm light and beeper	
Dimensions	W * H * D mm = 2800 * 2200 * 1400 mm	Weight: < 6500 kg

Power Module Configuration

Max 12 power module slots for the configuration	LCG1K0135G-M	DC2DC converter, 40kW/1kV, MPPT support	Max 2 * 40kWp solar power input support (optional)
	LCG1K0135G	DC2DC converter, 40kW/1kV	Max 10 * 40kW DC bus discharging power/EV charging power
	LRG1K0100G	AC2DC rectifier, 30kW/1kV	Max 2 * 30kW DC bus charging power
	LBG1K0120G	Bidirectional AC2DC inverter, 35kW/1kV	Max 2 * 35kW AC feedback power (optional)

Charging Dispenser

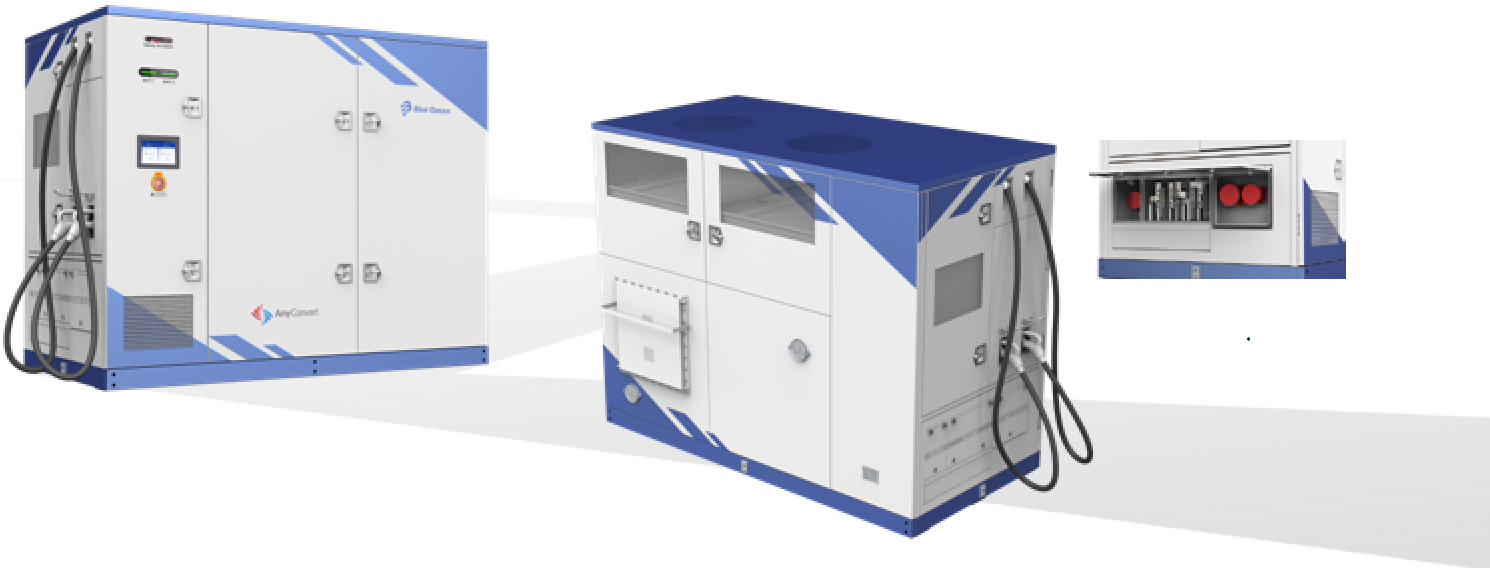
Charging Dispenser		EXP250-FDC	EXP375-FDC	EXP250-FSW-E	EXP375-FSW-E
DC output	Output charging cable	2	2	1	1
	Output DC voltage range	150V~1000V			
	Max DC output current	2 * 250A	2 * 375A	250A	375A
HMI	Tempered glass protective 7" capacitive Touch Screen LCD, RGB panel LED, E-STOP			Bottom RGB light, optional E-STOP panel	
Metering	DC energy meter, optional MIDI DC meter and Eichrecht certification configuration				
Monitor	OCPP 1.6J, support update to 2.0, P&C, smart charge and power management support, LAN 100M and LTE wireless modem support				
Cable management		Top assembly cable retractors with light bar		Bottom suspension cable retractor	
Payment		RFID, Mobil APP, POS (opt)		Mobile APP	
MBE function support		WEB side Maintenance Backend System enhances the operation and maintenance effectiveness			
Mechanical	Dimensions: W * H * D mm	500*1800*260/200kg/IP55/IK10		763*415*240/50kg/IP65/IK10	
Charging connector configuration		2 * CCS, or CCS + NACS		CCS or NACS	
Charging connector capacity		CCS2: 250A/375A, CCS1: 200A/300A, NACS: 200A/250A/300A/380A			
Noise		≤60dB @30°C @rated input and output full power, max≤65dB			

2 * 175kWH Liquid Cooled Battery Rack

Energy storage battery	Two groups 175kWh/285Ah liquid cooled battery rack. Bat volt: 537.6~691.2VDC, 1P dis/charge, max 1.1P discharge 10 min
High voltage control box	Two Infypower high voltage control box
Battery BMS	Infypower 2 level BMS structure: battery pack BSU + high voltage control box BMU
Thermal management	Two liquid chilling unit with 8kW refrigerating and 2.5kW heating capacity each, full Infypower design and manufacture
Fire suppression system	Completely submerged aerosol fire extinguishing system, water fire interface, emergency fan
Safety design	Explosion-proof board configuration and fire-proof spread structure design
Envirement sensor	Water sensor + temperature sensor + humidity sensor + door access seneor + smoke detector + variety of combustible gas detector
Protection class	IP67 battery pack, IP55 Battery cabinet area
Certification	Full TUV CE and US/CA

IHEBV470K7D-350LL3C Liquid Cooled BES Movable All in One Battery Charger

400kW Charging power/2 Charging Connectors/350kWh Battery Energy Storage Charger



Introduction

Off-grid and movable energy storage charging solution supports the temporary rescue charging service and off-grid power supply service. The energy storage charging system employs LFP battery for energy storage and helps balance the power supply and demand between the grid, battery and EV charging through the local and cloud EMS. Fully liquid cooling solution for battery groups can realize the 1P/1C discharge performance to meet the AC grid power limit and high power charging ability with energy cost efficiency and space saving, also can realize long life cycle. Full off-grid electric design and enhanced structural design can be suitable for long-term vibration transportation and temporary field charging.

Main Features

- Full liquid cooled design, from power module to the battery pack and charging cable
- Full electric safety design with the total electric isolation between the grid, battery, EV and solar power input
- Highly integrated all in one design, PV, ESS, EV charging and V2G functions
- Easy site-to-site transfer
- Friendly noise management
- 1P/1C fast charge and discharge
- Micro-grid with high charging power

Solution Values

- High power charging in off-grid status
- Temporary grid capacity expansion
- Movalble application
- Micro grid supply

Technical Parameters

IHEBV470K7D-350LL3C 470kW/350kWh BES Movable All in One Battery Charger

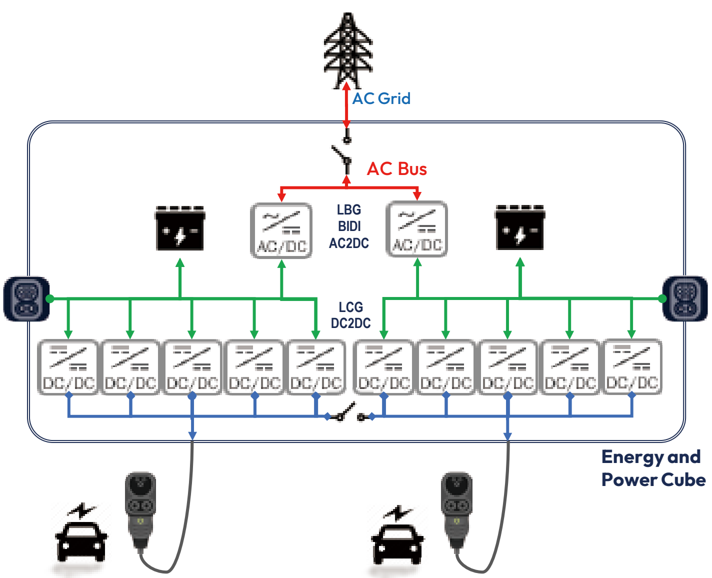
AC grid access	AC input voltage	45-65Hz/3-phase+(N)+PE/260Vac-530Vac
	AC max input power	70kW
	AC max output power	70kW
	Off-grid support	Yes
	Grid access interface	Industrial socket-inlets
External DC charger charging access	Access interface	2 CCS socket-inlets
	Access power	Max 200kW per channel, can charging in parallel
Electric vehicle charging	EV charging point	2 connectors
	EV charging power	400 kW
	EV charging power transfer	two connectors charge in parallel with half power or in series with full power
	EV charging voltage and current	150V~1000V, max 375A
Metering	AC grid side	1 AC energy meter, optional MID DC meter
	DC output side	2 DC energy meters, optional NMI Eichrecht certification configuration
HMI	Tempered glass protective 10" capacitive touch screen LCD, RGB panel LED, E-STOP batt alarm light and beeper	
Payment	RFID, optional credit card reader installation	
Monitor	OCPP 1.6J, support update to 2.0, P&C, smart charge and power management support, LAN 100M and LTE wireless modem support	
Noise	< 60dB	
Dimensions	W * H * Dmm = 2800 * 2200 * 1400mm	Weight: < 7000kg
Charging and billing modes	Auto full charge, charge time, amount, charge energy, charge end SOC	
Charging connector configuration	2 * CCS, or CCS + NACS	

Power Module Configuration

12 power module slots for the configuration	LCG1K0135G	DC2DC converter, 40kW/1kV	10 * 40kW ES Battery bus discharging/EV charging power
	LBG1K0120G	Bidirectional AC2DC inverter, 35kW/1kV	2 * 35kW AC power

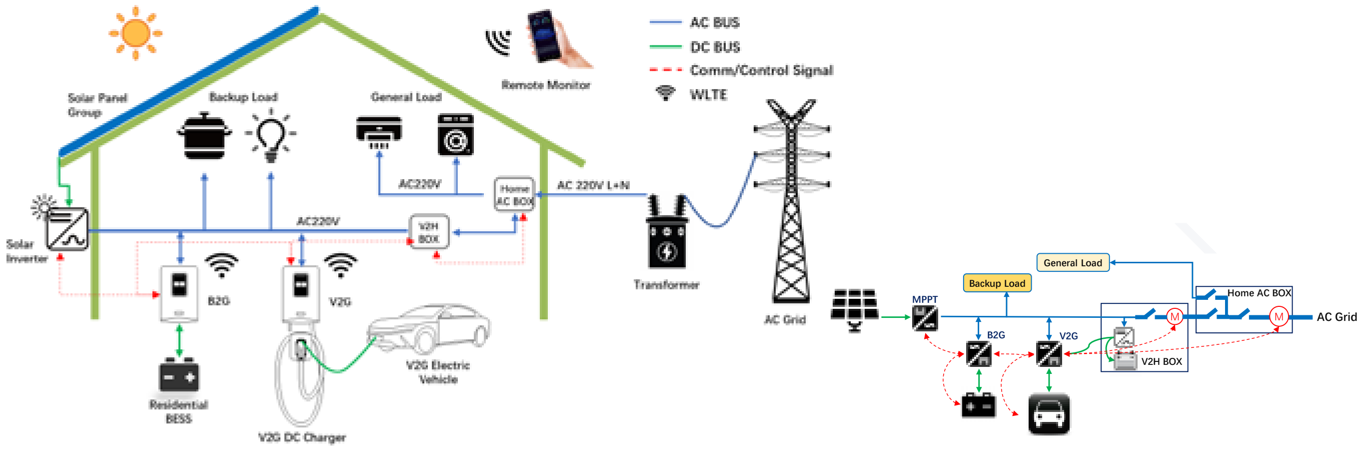
2 * 175kWH Liquid Cooling Battery Rack

Energy storage battery	Two groups of 175kWh/285Ah liquid cooling battery rack. Bat volt: 537.6~691.2VDC, 1 P/C dis/charge, max 1.1P/C discharge 10 min
High voltage control box	Two Infypower high voltage control boxes
Battery BMS	Infypower 2 level BMS structure: battery pack BSU + high voltage control box BMU
Thermal management	Two liquid cooling unit with 20kW refrigerating and 2.5kW heating capacity each, full Infypower design and manufacturing
Fire suppression system	Completely submerged aerosol fire extinguishing system, water fire interface, emergency fan
Safety design	Explosion-proof board configuration and fire-proof spread structure design
Envirement sensor	Water sensor + temperature sensor + humidity sensor + door access seneor + smoke detector + combustible gas detector
Protection class	IP67 battery pack, IP55 battery cabinet area
Charging power	External DC fast charger access, 2 * 200kW, or AC grid power access, 70kW
Discharging power	2 * 200kW



EXP07K1E-FSW

7kW 750V Single Phase V2G/V2H DC Wallbox Charger



Introduction

EXP07K1E is the bidirectional 7kW DC wall box for V2G/V2H and B2G/B2H applications, to connect the EV battery or the Residential energy storage battery to the single phase AC grid to realize the EV dis/charging V2G function and Home power backup V2H function. This V2G/V2H and B2G/B2H functions enable EV battery and residential battery to be used for grid energy storage, backup power supply and smart grid node.

Main Features

- Inside high frequency transformer isolation to enhance safety
- Residential using with low noise, <60dB, EMC class B
- Max efficiency is higher than 95.0%
- WEB/APP Maintenance Backend System enhances the operation and maintenance effectiveness
- Low standby power consumption, less than 30W
- IP65

Unique Function

- 7kW DC charger with AC single phase for CCS, CHAdeMO or GBT standard
- V2G, V2H, B2G, B2H configuration
- Optional V2H Box to realize the off grid power backup
- Wide DC voltage range from 300V to 750V
- One key to realize the manual charging and discharging

Application

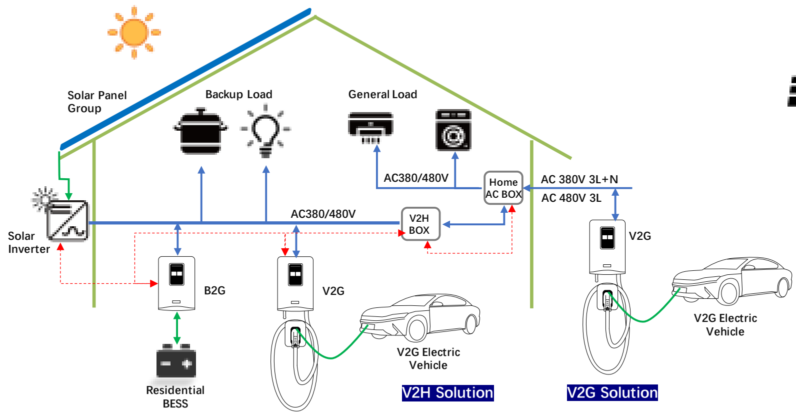
- V2G EV charger
- B2G Energy storage battery inverter
- V2H/B2H application with optional off grid V2H box
- Residential off grid backup power supply

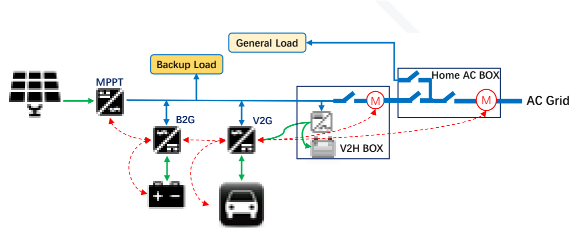
Technical Parameters

Environmental		Ambient temperature	-40°C~75°C, derating from 45°C
		Cooling	Fan cooling
		Altitude	2000m
Charging mode	AC input	Input voltage & current	230Vac, L+N+PE; 32A
		Input voltage/frequency range	100 Vac~264 Vac, 45 Hz~65 Hz
		Power factor	≥0.98 full-load output power of @50%~100%
		THD	≤5% full-load output power of @50%~100%
	DC output	Rated power	6.2kW
		Voltage and current range	300Vdc~750Vdc, 0~20.7A
		Efficiency (max)	≥95.0%
	V2G AC box	If there is no V2H box, need to select this box	AC energy meter, SPD, AC distribution
Charging and V2G mode convertion time (battery test)			100ms
V2G mode	DC input	DC input voltage and output power	See Table
		Max input current	22A
	AC output	Output AC voltage and output power	See Table
		Rated power and current	6.2kW/27A
		Output AC frequency	50 Hz/60 Hz
		THDi/output power factor	<5%/user setting, 0.8 leading~0.8 lagging
		Efficiency (max)	≥95.0%
	Off grid	Voltage accuracy and distortion	1% and <3%/off grid only support 230Vac
		Power factor	0.8~1
		Dynamic volt stability & recovery time	5% and 20ms
	V2H (optional)	V2H box	Bidirectional mid AC energy meter, SPD, AC distribution
			PV inverter connection, grid switch connector, backup 12V/5Ah battery
		Grid connection	Power switch in the main grid, backup grid and V2G charger by AC contactor
Protocol standard	OCPP 1.6J/OCPP 2.0, ISO15118/ISO15118-20, CHAdeMO V1.2, EVPOSSA		
B2G wallbox converter	Battery side: charge/discharge voltage 300~750V, max power 7kW, max current 23A		
	AC side: 100 Vac~264 Vac, 45 Hz~65 Hz, max current 32A, grid connection standard: same to V2G		
Control		Monitor interface	LAN 10M/100M and LTE wireless modem support
		Charging indication RGB light bar	Green-normal, yellow-charging, blue-discharging, red-failure
		HMI	Tempered glass protective 5" TFT touch screen LCD, RFID support
		Metering	Bidirectional mid AC energy meter in optional V2H box
		Hot key	One EPO key, one V2G action key
Reliability and safety	Protection	IP65, IK10 for cabinet and IK08 for LCD	
	EMC/safety	TUV CE, RED, EN61000-6-3/EN61000-6-1 Class B ; EN 61851-1/EN 61851-23/EN 61851-24	
Grid connection	Certification	VDE-AR-N 4105, EN50549, AS4777.2	
Mechanical		Dimensions/weight	612mm(H)×340mm(W)×187mm(D)/≤40kg (including the charging gun)
Charging cable configuration		40A/750V for CCS2, 37A/500V for CHAdeMO, 32A/750V for GBT	
Optional pedestal		Pedestal with cable retractors	

EXP11K1E-FSW

11kW 1000V Three Phase V2G/V2H DC Wallbox Charger





Introduction

EXP11K1E is the bidirectional 11kW DC wall box for V2G/V2H and B2G/B2H applications, to connect the EV battery or the Residential energy storage battery to the single phase AC grid to realize the EV dis/charging V2G function and Home power backup V2H function. This V2G/V2H and B2G/B2H functions enable EV battery and residential battery to be used for grid energy storage, backup power supply and smart grid node.

Main Features

- Inside high frequency transformer isolation to enhance safety
- Residential using with low noise, <60dB, EMC class B
- Max efficiency is higher than 96.0%
- WEB/APP Maintenance Backend System enhances the operation and maintenance effectiveness
- Low standby power consumption, less than 20W
- IP65

Unique Function

- 11kW DC charger with AC three phase for CCS, NACS, CHAdeMO or GBT standard
- V2G, V2H, B2G, B2H configuration
- Optional V2H Box to realize the off grid power backup
- Wide DC voltage range from 150V to 1000V
- One key to realize the manual charging and discharging

Application

- V2G EV charger
- B2G Energy storage battery inverter
- V2H/B2H application with optional off grid V2H box
- Residential off grid backup power supply

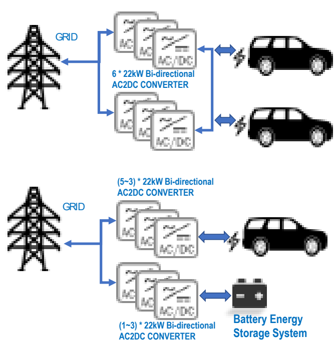
Technical Parameters

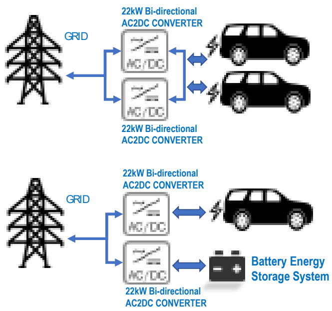
Environmental		Ambient temperature	-40°C~+75°C, derating from 45°C
		Cooling	Fan cooling
		Altitude	2000m
Charging mode	AC input	Input voltage & current	380/400/480Vac, 3L+(N)+PE; 16.7A
		Input voltage/frequency range	260Vac~530Vac, 45Hz~65Hz
		Power factor	≥0.98 full-load output power of @50%~100%
		THD	≤5% full-load output power of @50%~100%
	DC output	Rated power	11kW
		Voltage and current range	200Vdc~1000Vdc, 0~37A
		Efficiency (max)	≥96.0%
	AC box	If no V2H box, need this Box	AC energy meter, SPD, AC distribution
Charging and V2G mode convert time (battery test)			100ms
V2G mode	DC input	DC input voltage and output power	150Vdc~1000Vdc, 11kW
		Max input current	37A
	AC output	Output AC voltage and output power	From 320 to 530Vac, ouput power is 11kW
		Rated power and current	11kW/16.7A
		Output AC frequency	50Hz/60Hz
		THDi/output power factor	<5%/user setting, 0.8 leading~0.8 lagging
		Efficiency (max)	≥95.5%
	Off grid	Voltage accuracy and distortion	1% and <3%/off grid only support 380Vac
		Power factor	0.8~1
		Dynamic volt stability & recovery time	5% and 20ms
	V2H box (optional)	V2H box	Bidirectional Mid AC energy meter, SPD, AC distribution
			PV inverter connection, grid switch connector, backup 24V/5Ah battery
		Grid connection	Power switch in the main grid, backup grid and V2G charger by AC contactor
Protocol standard	OCPP 1.6J/OCPP 2.0, ISO15118/ISO15118-20, CHAdeMO V1.2, EVPOSSA, optional EEBUS		
B2G wallBox converter	Battery side: charge/discharge voltage 150~1000V, max power 11kW, max current 37A		
	AC side: 320 Vac~530 Vac, 45 Hz~65 Hz, max current 16.7A, grid connection standard: same to V2G		
Control		Monitor interface	LAN 10M/100M and LTE wireless modem support
		Charging indication RGB light bar	Green-normal, Yellow-charging, Blue-discharging, Red-failure
		HMI	Tempered glass protective 5" TFT Touch Screen LCD, RFID support
		Metering	Bidirectional Mid AC energy meter in optional V2H box
		Hot key	One EPO key, one V2G action key
Reliability and safety	Protection	IP65, IK10 for cabinet and IK08 for LCD	
	EMC/safety	TUV CE, RED, EN61000-6-3/EN61000-6-1 Class B ; EN 61851-1/EN 61851-23/EN 61851-24; TUV US+CA, UL2202	
Grid connection	Certification	VDE-AR-N 4105, EN50549, AS4777.2, UL1741SB: 2021, CSA C22.3 No. 9 and CSA C22.2 No. 107.1	
Mechanical		Dimensions/weight	705mm(H)×405mm(W)×230mm(D)/≤45 kg (including the charging connector)
Charging cable configuration		40A/1000V for CCS1/2, 37A/500V for CHAdeMO, 40A/1000V for GBT, 50A/1000V for NACS	
Optional pedestal		Pedestal with cable retractors	

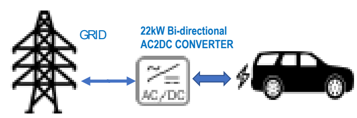
V2G Charging Solution

22kW/44kW/132kW V2G DC Fast Charger

■ EXP132K3E-FD 132kW V2G charger ■EXP44K3E-FDW 44kW V2G charger ■EXP22K3E-FSW 22kW V2G charger







Introduction

The V2G charger ensures the power supply and demand in good balance between the grid and EV battery by taking the EV battery as an energy storage device and through the help of local or remote Energy Management System. It also supports flexible PV energy access to optimize grid peak-valley electricity usage, supplement the grid capacity and provide backup power supply. It can be a core node to smart grid or micro grid access and also an important supplement to the user-side energy storage system.

Main Features

- Flexible change of the system configuration, capacity and the direction of power flow, customization support
- WEB/APP Maintenance Backend System enhances the operation and maintenance effectiveness
- Total electric insolation between the grid, battery and EV
- Fully compatible with different system configurations
- Global V2G standard support
- Unified EMS strategy

Solution Values

- Grid peak valley electricity using
- Grid capacity supplement
- Grid quality and safety improvement
- Supplement to the user-side energy storage

Technical Parameters

		EXP22K3E	EXP44K3E	EXP132K3E
AC grid access	AC mode	45-65Hz/3-phase+(N)+PE/260Vac-530Vac		
	Max AC feedback power	22 kW	44 kW	132 kW
	Power factor	>0.99		
	Total harmonic current	<3% (rated input)		
EV charging	Max charging power	22 kW	44 kW	132 kW
	Charging power switch	Single charging connector	Charging in parallel with half power or in series with full power between 2 charging connectors	
	Charging voltage	150V~1000V		
EV discharging	Max V2G feedback power	22 kW	44 kW	132 kW
	V2G protocol standard	CCS: DIN70121, IEC15118/-2/-20, CHAdeMO V1.2, EVPOSSA		
Optional battery energy storage access	Battery voltage	/	300~1000V	
	Max dis/charging power	/	22 kW	66kW configuration
	BES BMS access	CAN communication		
Metering	AC Grid side (optional)	One bidirectional AC energy meter		
	Charging side	Bidirectional DC energy meter	Two bidirectional DC energy meter	
Dimensions	W * H * D mm	610 * 640 * 270 mm	705 * 1100 * 240 mm	700 * 1750 * 750 mm
Weight	kg	65 kg	120 kg	250 kg
Protection class		IP55/IK10	IP55/IK10	IP55/IK10
Thermal management		Air cooled		
Ambient temp	-30 ~ +70°C, full power output below 50°C, power derating 5%/°C above 50°C			
EMC/safety	TUV US+CA, UL2202, TUV CE/RED EN62909, EN61000-6-3/EN61000-6-1 Class A; EN 61851-1/EN 61851-23/EN 61851-24			
Grid connection	VDE-AR-N 4105, EN50549, UL1741SB			

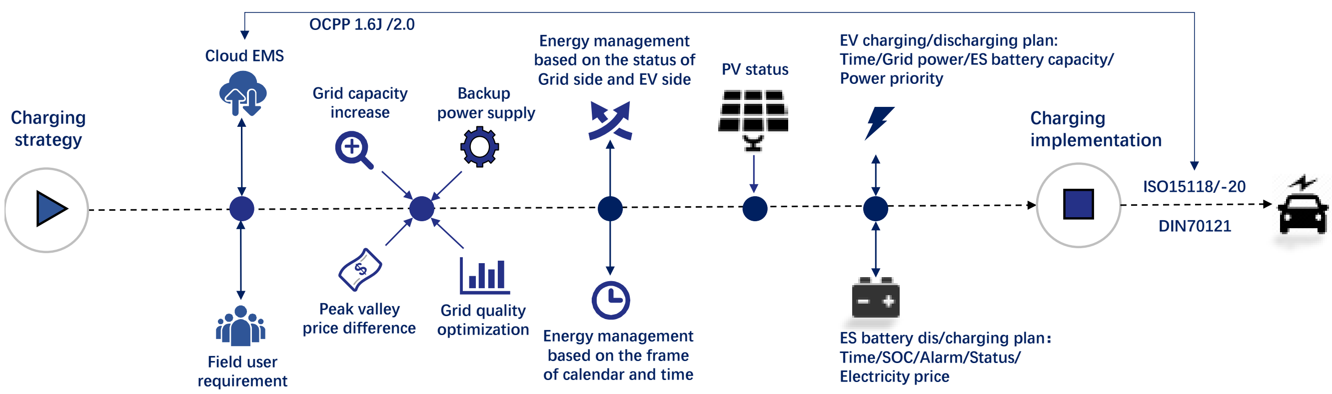
Configuration

Power module configuration	1 * 22kW power module	2 * 22kW power modules	6 * 22kW power modules
Charging connector configuration	1 CCS +1 CHAdeMO or 2 CCS or 1 CCS or 1 CHAdeMO		

Function and Interface

HMI	Tempered glass protective 7" TFT touch screen LCD, RFID, RGB panel LED, POS (opt)
Back-end platform	OCPP 1.6J, support firmware update to 2.0x, OTA support, P&C/smart charge/power management support
MBE function support	WEB side Maintenance Backend System enhances the operation and maintenance effectiveness
Charging management	Pedestal with two cable retractors and the light bar

V2G Charging System Running Strategy



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E-mail: info@unicom-bass-systems.com

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